CONGENITAL GLAUCOMAS

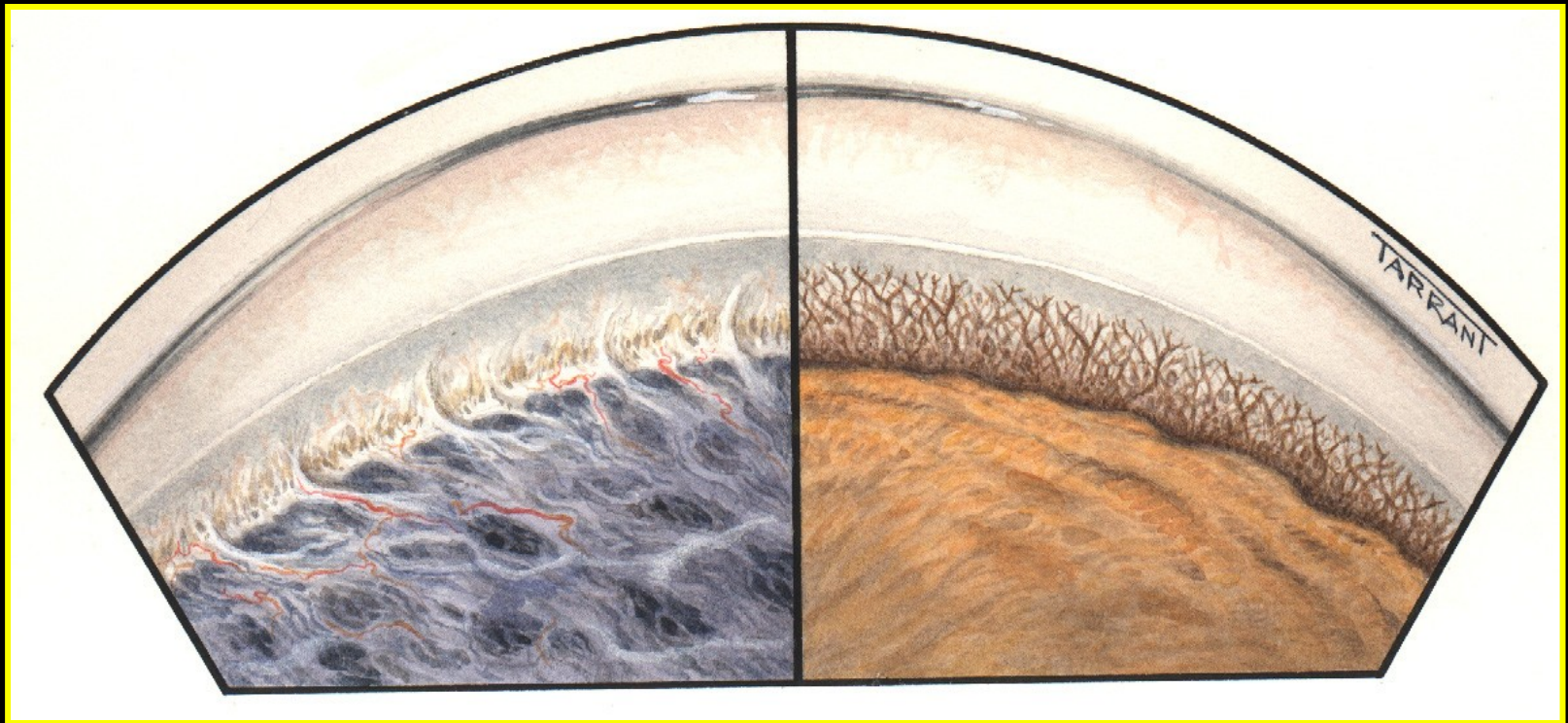
1. **Primary**

2. **Iridocorneal dysgenesis**
   - Axenfeld-Rieger anomaly
   - Peters anomaly
   - Aniridia

3. **In phacomatoses**
   - Sturge-Weber syndrome
   - Neurofibromatosis - 1

Primary congenital glaucoma

- 1:10,000 births, 65% boys
- Most sporadic - 10% autosomal recessive
- Absence of angle recess with iris inserted directly into trabeculum

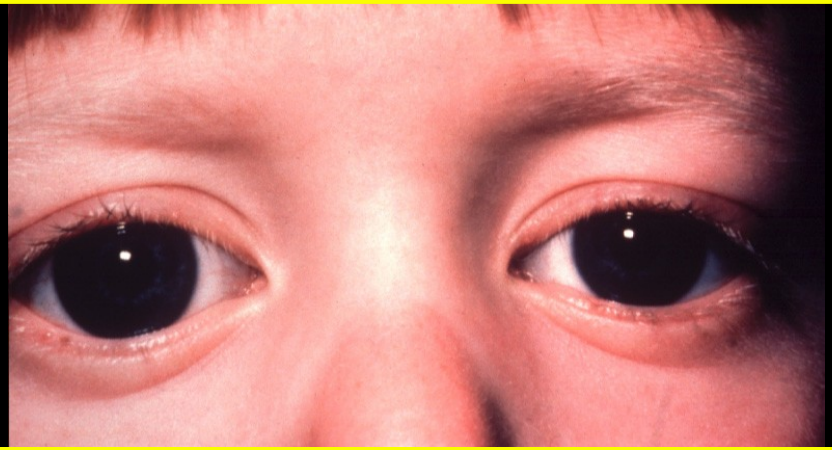


Flat iris insertion	Concave iris insertion

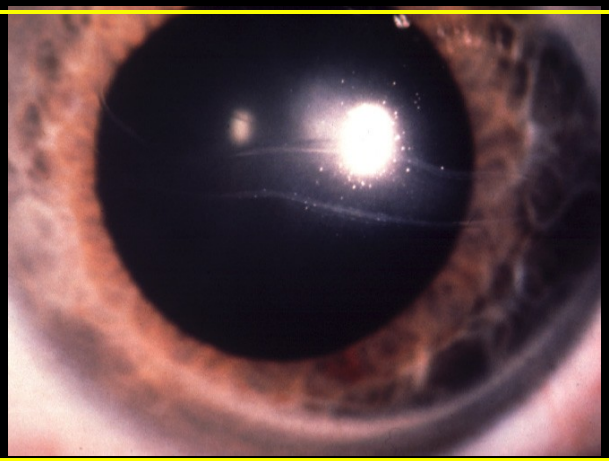
Clinical features of primary congenital glaucoma

- Depend on age of onset
- Bilateral in 75% but frequently asymmetrical

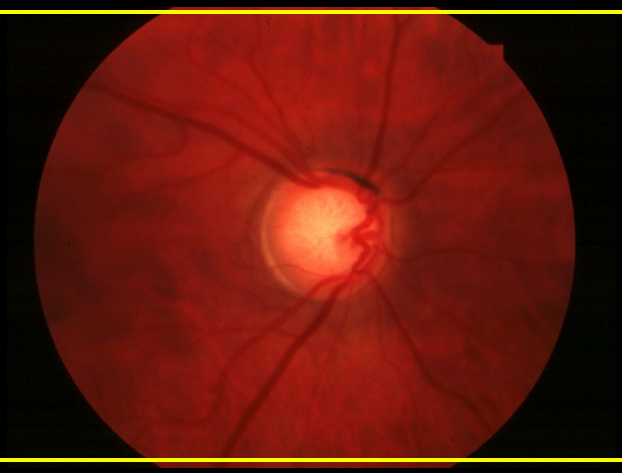
Corneal oedema associated with lacrimation and photophobia



Buphthalmos if IOP becomes elevated prior to age 3 years.

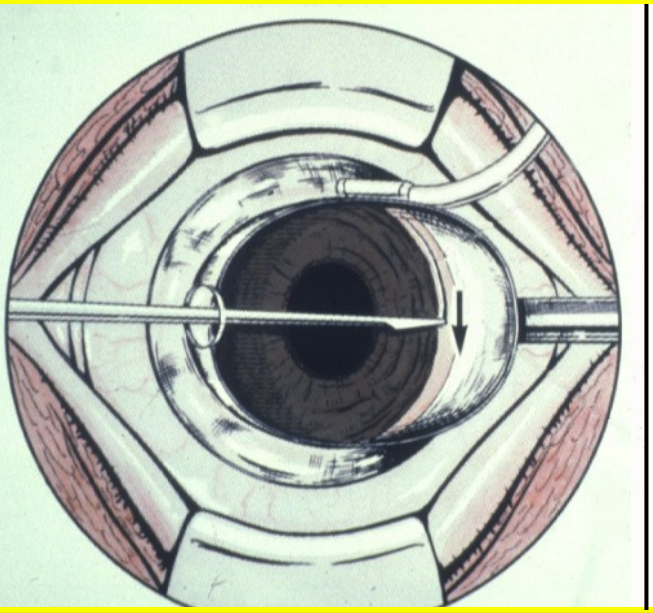


Breaks in Descemet membrane



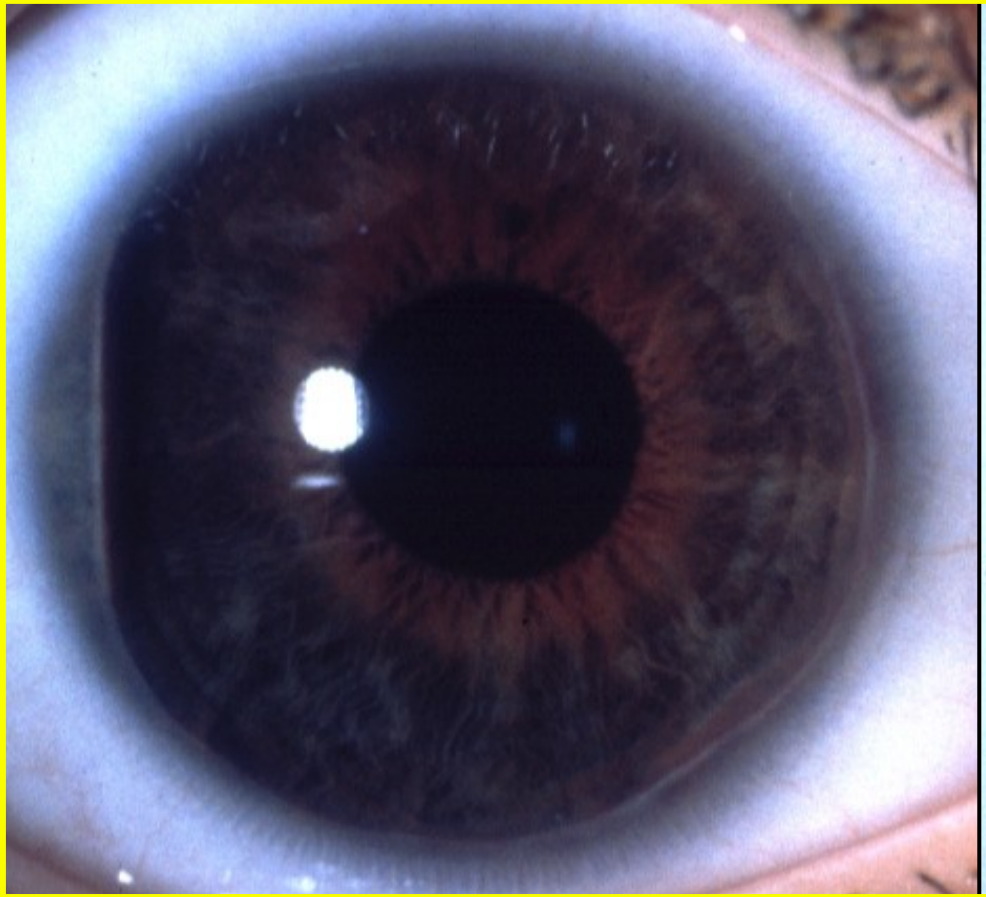
Optic disc cupping

Management of primary congenital glaucoma

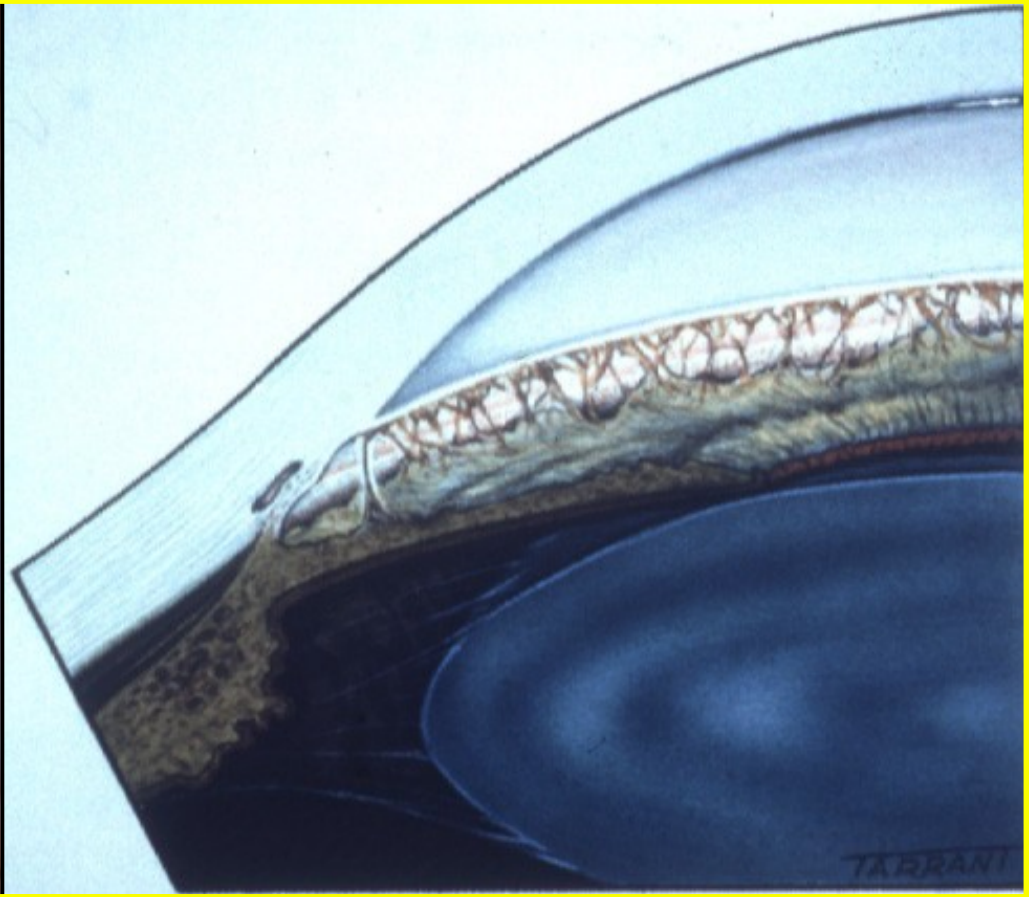
Measurement of IOP and corneal diameters	Goniotomy	Trabeculotomy
		

Axenfeld anomaly

- Bilateral but asymmetrical
- Glaucoma is uncommon



Posterior embryotoxon

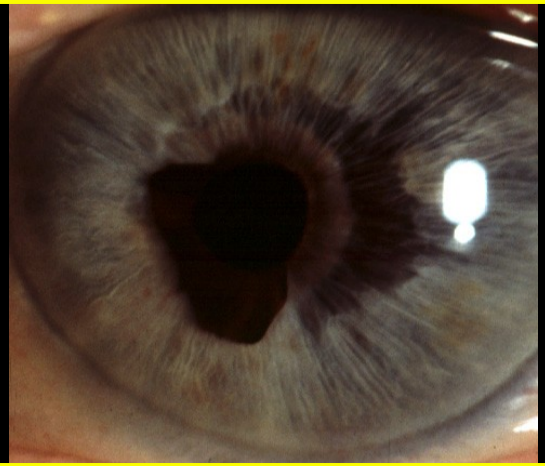


Attached strands of iris to posterior embryotoxon

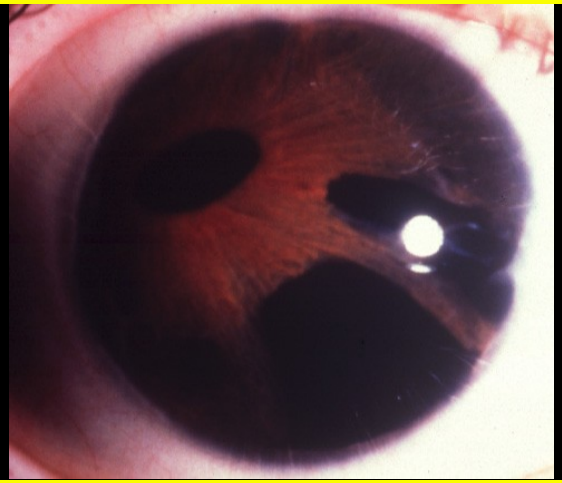
Rieger anomaly

- Autosomal dominant
- Bilateral but asymmetrical
- Glaucoma in 50%

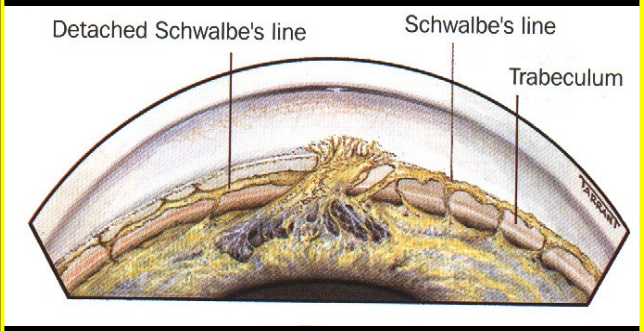
Stromal hypoplasia and corectopia



Ectropion uveae



Full-thickness iris atrophy



Angle anomalies

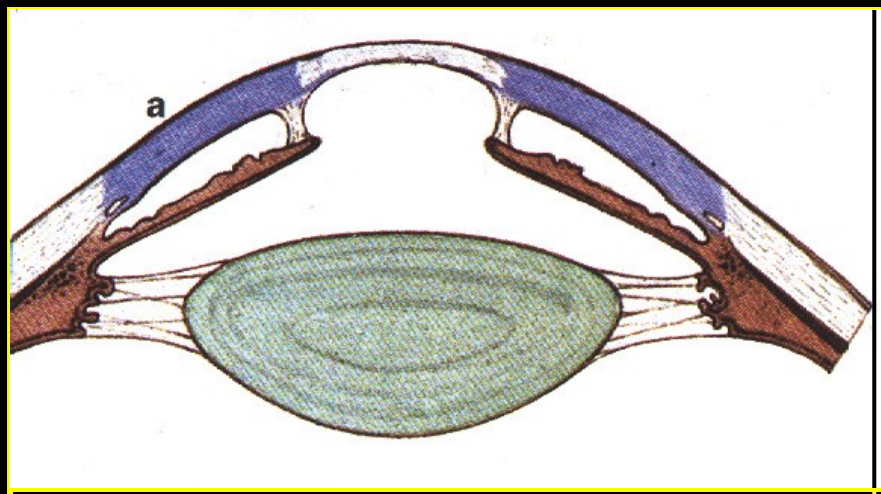
Rieger syndrome

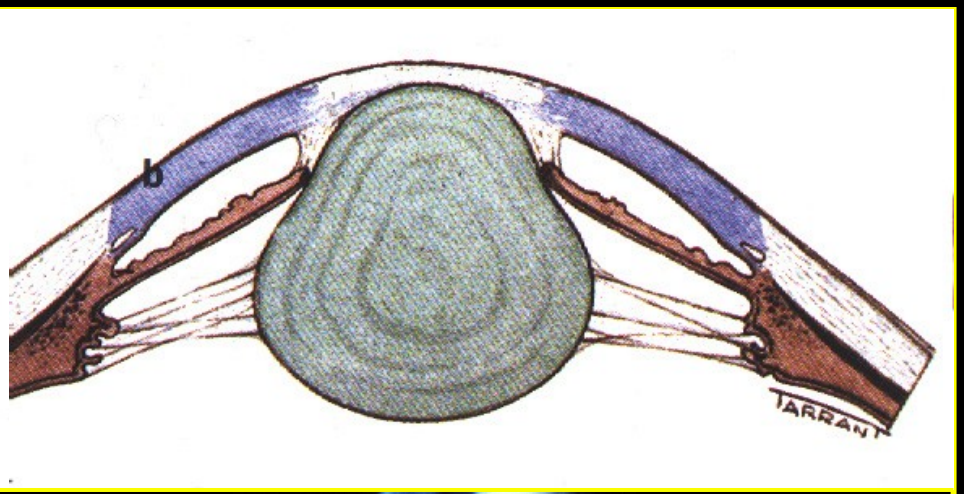
Rieger anomaly

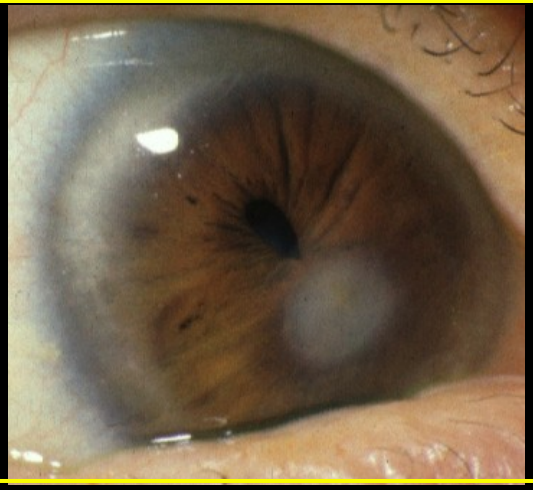
Dental and facial anomalies

Peters anomaly

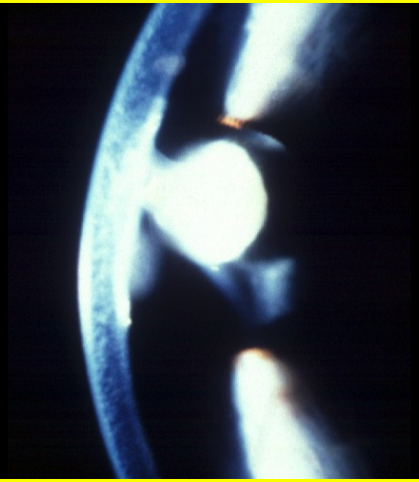
- Usually sporadic
- Bilateral in 80%
- Glaucoma in 50%







Corneal opacity with iris adhesions



Corneal opacity with lenticular adhesions

Systemic Implications of Aniridia

AN-1 - 85%

- Autosomal dominant
- Isolated

AN-2 (Miller syndrome) - 13%

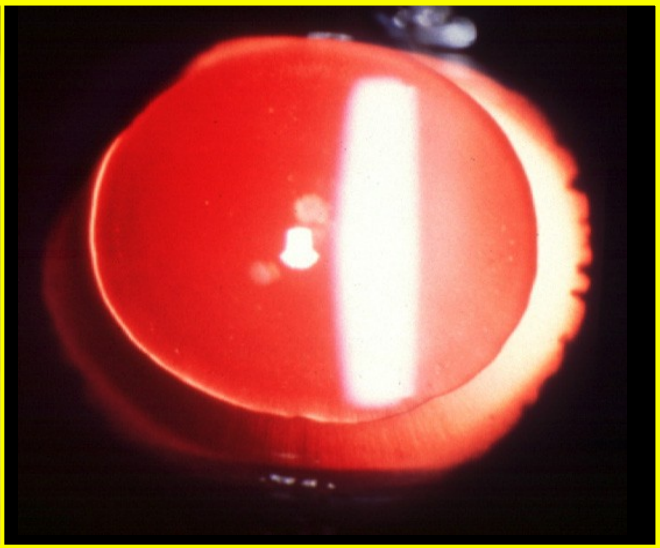
- Deletion of short arm of chromosome 11
- Wilm tumour, genitourinary anomalies and mental handicap

AN-3 (Gillespie syndrome) - 2%

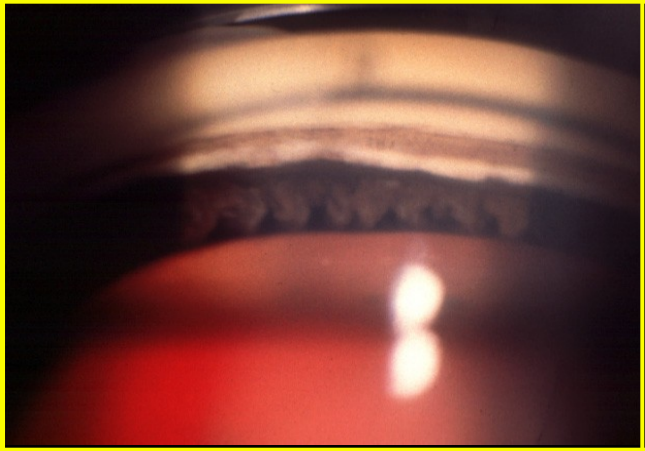
- Autosomal recessive
- Mental handicap and cerebellar ataxia

Signs of aniridia

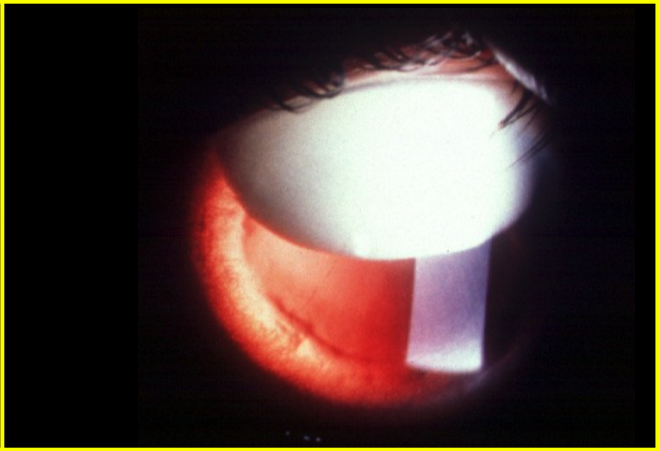
Partial absence



Subtotal absence



Synechial angle-closure glaucoma in 75%

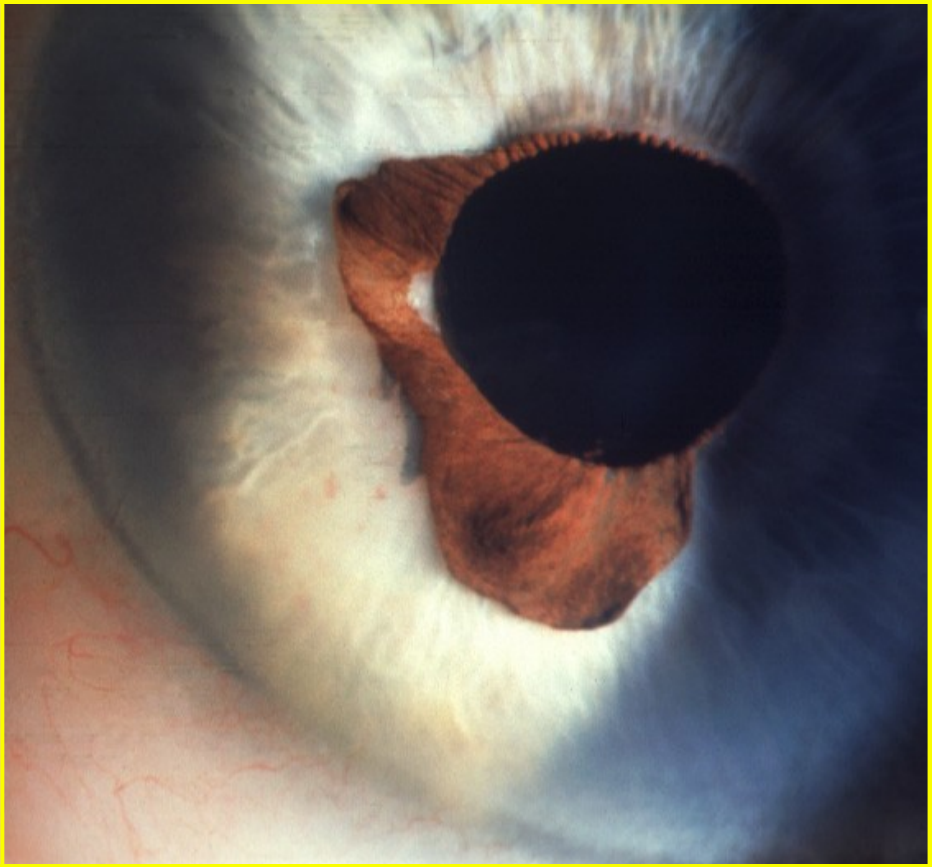


Occasional cataract and lens subluxation

Glaucoma in Sturge-Weber syndrome

Glaucoma	Causes
• Glaucoma in 30% • Ipsilateral to facial haemangioma • Buphthalmos in 60%	• Caused by raised episcleral venous pressure associated with episcleral haemangioma • Angle anomaly may also be responsible

Glaucoma in neurofibromatosis - 1

Glaucoma	Causes
	
• Glaucoma is ipsilateral to neurofibroma of upper eyelid in 50% of cases	• Caused by angle anomaly with or without ectropion uveae • Angle neurofibroma may also be responsible